

A054 v0.5.0: E013 Provider-Convergent State-Selection Falsifier

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1 Identification

Catalog ID A054; release v0.5.0; SST Falsifier Framework v1.0.6, profile `multilibrary_gpu`. This is an E013 cross-falsifier member at stage 30.

2 Question

Can distinct current E011/PKLSA provider representatives of the same topology be mapped by one target-free variational selector to a common stationary and locally stable finite-core state?

3 Hypotheses

The null states that provider dependence survives target-free state selection or that selected stationary states fail local stability. The alternative requires provider-convergent stationary and stable endpoints without topology/provider-specific retuning.

4 Assumptions

E013 fixes the current carrier population. The selector is a variational existence/selection test, not a claim that inviscid Euler flow is dissipative. Physical stability is tested separately after stationarity. Legacy falsifier results are excluded as evidence.

5 Symbols

$\hat{H}_a$ is the reach-normalized regularized line-energy kernel, q Bishop-frame deformation coordinates, σ the normalized energy-gradient stationarity score, and K_H the endpoint Hessian.

6 Equations

Machine-readable science contract

Research question On one E013-frozen population of current E011 provider anchors, can a fixed target-free variational state-selection procedure map distinct source representatives of the same topology to a common stationary and locally stable finite-core state, without using legacy falsifier results?

Objective Re-run state selection from source-native bytes for exactly the E013 carriers; quantify stationarity, topology preservation, provider-endpoint convergence, local Hessian stability and nonlinear reduced ringdown; freeze only provider-converged stable endpoints as a future A057 handoff.

H_0 H0: provider-distinct current PKLSA/SKLSA representatives do not converge under the pre-registered selector to a reproducible stationary and stable topology-level state beyond numerical/source variation.

H_1 H1: the same preregistered selector, without topology/provider-specific retuning, yields provider-convergent stationary and locally stable endpoints for a reproducible subset of topologies.

Registered equations

E1: Regularized finite-core comparison Hamiltonian

$$\hat{H}_a = \sum_{i,j} \oint \oint \frac{d\hat{\mathbf{X}}_i \cdot d\hat{\mathbf{X}}_j}{\sqrt{\|\hat{\mathbf{X}}_i - \hat{\mathbf{X}}_j\|^2 + \hat{a}^2}} \quad (1)$$

Dimensional check: All lengths are reach-normalized; the kernel is dimensionless.

E2: Bishop-Fourier reduced deformation

$$\delta \mathbf{X}(q) = \sum_{\mu} q_{\mu} \mathbf{B}_{\mu}, \quad \mathbf{B}_{\mu} \perp \mathbf{t} \quad (2)$$

Dimensional check: In reach-normalized coordinates both q and B are dimensionless.

E3: Stationarity score

$$\sigma = \frac{\|\nabla_q \hat{H}_a\|_2}{\max(|\hat{H}_a|, \epsilon)} \quad (3)$$

Dimensional check: Ratio of dimensionless quantities is dimensionless.

E4: Target-free variational selector with backtracking

$$q_n = -\eta_n \frac{\nabla_q \hat{H}_a}{\|\nabla_q \hat{H}_a\|}, \quad \mathbf{X}_{n+1} = \mathcal{N}[\mathcal{R}(\mathbf{X}_n + \delta \mathbf{X}(q_n))] \quad (4)$$

Dimensional check: All operations occur in reach-normalized coordinates.

E5: Accepted-step integrity conditions

$$\hat{H}_{n+1} \leq \hat{H}_n, \quad d_{\min, n+1} \geq f_c d_{\min, 0}, \quad |\Delta Lk| \leq \varepsilon_{Lk} \quad (5)$$

Dimensional check: Energy comparison is dimensionless; distance is compared as a ratio; linking is dimensionless.

E6: Cross-provider endpoint agreement

$$CV_H(K) = \frac{s(H_K^*)}{|\bar{H}_K^*|}, \quad \Delta_R(K) = \frac{R_{\max}^* - R_{\min}^*}{|\bar{R}^*|} \quad (6)$$

Dimensional check: Coefficient of variation and relative spans are dimensionless.

E7: Endpoint local stability Hessian

$$K_H = \nabla_q^2 \hat{H}_a \Big|_{q=0} \quad (7)$$

Dimensional check: Dimensionless in normalized coordinates.

E8: Nonlinear reduced ringdown

$$\ddot{z} = -\partial_z U(z), \quad U(z) = \hat{H}_a(z e_{\min}) \quad (8)$$

Dimensional check: Normalized time and coordinates make the reduced test dimensionless.

E9: State-selected handoff predicate

$$Q_K = I[\text{provider consensus}]I[\sigma \leq \sigma_{\max}]I[K_H \succeq 0]I[\text{ringdown bounded}] \quad (9)$$

Dimensional check: Boolean predicate.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol/implementation and exact E013/E011 source-byte binding.
2. **S2** [G1; equations E1]: Reload only E013-frozen carriers and recheck geometry hashes.
3. **S3** [G2; equations E1, E4, E5]: Validate rigid-energy invariance, selector monotonicity and topology-proxy mechanics.
4. **S4** [G3; equations E2, E3, E4, E5]: Run the fixed variational selector independently on every current carrier.
5. **S5** [G4; equations E3, E4]: Evaluate fixed-selector transfer on registered confirmation topologies.
6. **S6** [G5; equations E1]: Certify the common energy kernel against C++/OpenMP FP64.
7. **S7** [G6; equations E1]: Record optional GPU/DD32 framework screening without promotion.
8. **S8** [G7; equations E6]: Compare independently sourced endpoints topology-by-topology.
9. **S9** [G8; equations E7, E8]: Qualify stationary endpoints by Hessian and nonlinear ringdown.
10. **S10** [G10; equations E6]: Test whether initially provider-sensitive energy kernels collapse after selection.
11. **S11** [G11; equations E5]: Audit clearance/linking topology proxies over all paths.
12. **S12** [G12; equations E9]: Freeze only provider-converged stable endpoints as a future A057 handoff.
13. **S13** [G22; equations E3, E6, E7, E9]: Assign the blind overall state-selection classification under diagnostic continuation.
14. **S14** [G9; equations E9]: Reserve framework reveal integrity gate; v0.5.0 contains no mass target reveal.

7 Algorithm

Every E013 carrier is reloaded from exact source bytes, resampled, reach-normalized and independently relaxed by the same Bishop-Fourier reduced steepest-descent selector with monotone backtracking. Accepted steps must preserve a nonlocal-clearance floor and pairwise linking within tolerance. The endpoint is then re-tested for stationarity, provider convergence, Hessian stability and nonlinear reduced ringdown.

8 Sources

E011-v0.3.0 is the static-seed/provider authority; E013-v0.1.1 freezes the common carrier population; E010-v0.3.1 only reloads the exact source-native geometry bytes. No old A054 output is a science input.

9 Gates and diagnostic continuation

Only G0 is a hard prerequisite. Scientific failures do not suppress later diagnostics. Soft dependency caveats prevent a numerical diagnostic from being promoted beyond its support.

10 Numerics

BASIC, FULL and CERTIFY increase reduced basis richness, resolution, iteration budget and stationarity strictness. The same global selector policy is used for every carrier; no provider/topology-specific optimization hyperparameter is fitted.

11 Backends

Python/NumPy FP64 is REFERENCE. C++/pybind11 OpenMP FP64 certifies the common line-energy kernel at relative-L2 $\leq 10^{-10}$. Framework SYCL/DD32 remains SCREENING_ONLY.

12 Blindness and provenance

There are no protected mass targets in v0.5.0. Blindness is used to lock the protocol and to exclude legacy scientific outputs. E013 carrier identities and provider groups remain visible because provider convergence is the object of the test.

13 Statistics

Primary statistics are carrier stationarity success fraction, provider endpoint energy CV, ropelength relative span, absolute-writhe span, source-to-endpoint provider-CV collapse, Hessian sign and nonlinear ringdown amplitude/energy drift.

14 Results

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	28bd86860879fa370e1a573d60a915af61f312dfa36f7f04298d78099c2bb2bc
Framework schema	SST-FALSIFIER-PROTOCOL-2

Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	a3ef7af4fe2b28e051942e5a9a979b3c73081d69aa5bebb7a6e4ebbe1df7302a

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen v0.5.0 protocol intact and runtime/implementation provenance self-consistent?
G1	PASS	Is the run exactly bound to the E013 common-carrier manifest, with all E011 source ge
G2	PASS	Do rigid-objectivity, null geometry, selector monotonicity and topology-proxy controls p
G3	FAIL	Can the fixed target-free variational selector drive a preregistered fraction of current EC
G4	FAIL	Does the same frozen selector generalize without refit to registered confirmation topolog
G5	PASS	Does C++/OpenMP FP64 reproduce Python/NumPy FP64 for the state-selection ener
G6	PASS	If available, does framework GPU/DD32 screening evidence remain consistent without l
G7	FAIL	Do independently sourced provider anchors of the same topology converge to the same
G8	FAIL	Are stationarity-qualified endpoints locally stable and bounded under the preregistered
G10	FAIL	For source-energy-sensitive topologies, does state selection collapse provider spread towa
G11	PASS	Are clearance and link-drift topology proxies preserved along every completed state-sele
G12	FAIL	Can at least one provider-converged stationary stable state be frozen as an A057-compa
G22	FAIL	What is the overall state-selection evidence classification under diagnostic continuation?
G9	DEFERRED	After a completed FULL/CERTIFY blind run, do reveal and commitment checks verify

Backend manifest

```
{
  "cpp_certification": {
    "authority": "CERTIFICATION",
    "gate_status": "PASS"
  },
  "diagnostic_continuation": true,
  "gpu_screening": {
    "authority": "SCREENING_ONLY",
    "gate_status": "PASS"
  },
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scientific_gate_dependencies": "soft; see DEPENDENCY_CAVEATS.json"
}
```

Source manifest

```
{
  "manifest_sha256": "44be8e3e1053c49ef294c29e9946bab0e8aec78241e0bd17c936446578afee49",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
```

```

    "evidence_class": "independent_source",
    "exists": true,
    "independence_family": "E011_SKLSA_v0.3.0",
    "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E011_sklsa_s",
    "provenance_family": "E011_SKLSA_STATIC_READY",
    "registry_hint": {
        "catalog_id": "E011",
        "hits": []
    },
    "role": "static_seed_qualification_authority",
    "sha256": "eda7cdcddaf851d6e83b225df5182634a70b9678bfe3d3a0f338a28c2a0e8668",
    "source_id": "S_E011_STATIC_INDEX",
    "version": "v0.3.0"
},
{
    "catalog_id": "E011",
    "evidence_class": "independent_source",
    "exists": true,
    "independence_family": "E011_SKLSA_v0.3.0",
    "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E011_sklsa_s",
    "provenance_family": "E011_SKLSA_STATIC_READY",
    "registry_hint": {
        "catalog_id": "E011",
        "hits": []
    },
    "role": "provider_anchor_authority",
    "sha256": "999739dad02aa2b4dbc6c00411c9e21cfc59252de5daff65eb6471a168782d7d",
    "source_id": "S_E011_PROVIDER_ANCHORS",
    "version": "v0.3.0"
},
{
    "catalog_id": "E013",
    "evidence_class": "simulation",
    "exists": true,
    "independence_family": "E013_protocol",
    "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E013_PKLSA_C",
    "provenance_family": "E013_PKLSA_Cross_Falsifier",
    "registry_hint": {
        "catalog_id": "E013",
        "hits": []
    },
    "role": "cross_falsifier_common_carrier_protocol",
    "sha256": "162a65d09d431ebee4943a1f348770f6f7c442d257815f96d6bcb6cf2377f187",
    "source_id": "S_E013_CAMPaign_CONFIG",
    "version": "v0.1.1"
},
{
    "catalog_id": "E010",
    "evidence_class": "simulation",
    "exists": true,
    "independence_family": "E010_implementation",
    "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E010_pklsa_p",
    "provenance_family": "E010_PKLSA_v0.3.1",

```

```

    "registry_hint": {
      "catalog_id": "E010",
      "hits": []
    },
    "role": "source_native_geometry_parser",
    "sha256": "597d5e3b06b0a91884edd2a7ffa99b2c7160940c68c9da39addb0d0c32796d26",
    "source_id": "S_E010_LOADER",
    "version": "v0.3.1"
  }
],
"workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

15 Interpretation

A stationary point is not a stable state merely because $\nabla H \simeq 0$. Provider convergence and positive local stability are independent requirements. Only topologies satisfying both plus bounded ringdown are eligible for the state-selected handoff.

16 Reproducibility

Run `run_00_install.cmd`, then execute through E013 with `run_all_cross_falsifier.cmd` BASIC/FULL/CERTIF
The first A054 run automatically freezes the create-once protocol.

References

- [1] H. K. Moffatt, “The degree of knottedness of tangled vortex lines,” *J. Fluid Mech.* 35, 117–129 (1969). DOI: 10.1017/S0022112069000991.
- [2] P. G. Saffman, *Vortex Dynamics*. Cambridge University Press (1992). DOI: 10.1017/CBO9780511624063.
- [3] R. L. Bishop, “There is More than One Way to Frame a Curve,” *Amer. Math. Monthly* 82, 246–251 (1975). DOI: 10.1080/00029890.1975.11993807.